

$$H_0: \mu_d = 0 \quad H_a: \mu_d \neq 0$$

$$t^* = \frac{0.06 - 0}{\frac{0.114}{\sqrt{15}}} \approx 1.176$$

$$t_{4; 0.025} = 2.776$$

rejection region: $t > |2.776| \Rightarrow$ Not reject

DATA
SET

DS1026

26. Memory Experiments Twenty students participated in an experiment in which they were first asked to recall a list of 25 words. They were then given a second list, and asked to form images of the words as they were read. The number of words recalled for each student is given in the table.

Student	With Imagery	Without Imagery
1	20	15
2	24	15
3	20	15
4	18	9
5	22	16
6	19	11
7	20	8
8	19	11
9	17	7
10	21	9
11	17	8
12	20	16
13	20	10
14	16	12
15	24	7
16	22	9
17	25	21
18	21	14
19	19	12
20	23	13

Does it appear that the average recall score is higher when imagery is used?

$$\bar{D} = 10.25$$

$$S_D = 4.051$$

$$H_0: D_0 = 0, H_a: D_0 > 0, \alpha = 0.05$$

$$t_{STAT} = \frac{\bar{D} - D_0}{\frac{S_D}{\sqrt{n}}} \sim t_{19}$$

$$t^* = \frac{10.25}{\frac{4.051}{\sqrt{20}}} = 11.316$$

$$\text{rejection region: } t > 1.729$$

$$t_{19; 0.05} = 1.729$$

$$t^* = 11.316 > 1.729 \Rightarrow \text{reject } H_0$$

Conclusion: average recall score is higher